PARTS AND REFERENCE SHEET

VERIFY EVERY PART NUMBER BY VIN BEFORE YOU BUY

A four-sensor California 4.0L and a two-sensor federal 4.0L do not share oxygen-sensor part numbers or harness lengths. Count the sensors first. The position is part of the order: front or rear, before or after the mini-cat.

Oxygen sensors — 2004 4.0L numbers that were actually published

Source: JeepSpecs WJ oxygen-sensor guide, a compiled reference, not a dealer VIN lookup. The page lists four numbers for the 2004 4.0L and says federal and California are distinguished in the original table by color. That color key did not survive as a reliable position map. Hand all four to the counter with the VIN. Do not buy a set.

Published number	How the guide groups it	Status
56028995AA	2004 4.0L, before catalyst, one of two	CANDIDATE Position not proven here
56028994AA	2004 4.0L, before catalyst, one of two	CANDIDATE Position not proven here
56028997AA	2004 4.0L, after catalyst, one of two	CANDIDATE Position not proven here
05139021AA	2004 4.0L, after catalyst, one of two	CANDIDATE One owner reported this box as the rear downstream, sensor body stamped 56044598AA. Single report. Not a confirmation.

Numbers supplied for this job that did not check out

NTK 23142. Catalog listings found for this number are other Jeeps: 2001–2003 Wrangler / TJ downstream, Bank 1 Sensor 2, and 2002–2003 Liberty Bank 2 Sensor 2. Not a 2004 Grand Cherokee 4.0L listing. NOT CONFIRMED Do not buy it unless the dealer or the NTK catalog for this VIN and this bung returns 23142.

Mopar 56029001AB. No fitment for this vehicle was found. Nearby real numbers (56028994AA, 56029049AA, and the four above) are different parts. NOT CONFIRMED Do not order 56029001AB from this sheet.

If the heater test says the sensor is bad

Order one sensor, for Bank 2 Sensor 2, rear pipe after the rear mini-cat, by VIN. Prefer new Mopar or NTK. NTK supplied many Mopar sensors of this era. LIKELY from the parts trade, not a factory letter. Install torque cited in the WJ guide: 22 ft-lb (30 N·m). Do not add anti-seize. Do not install it into an oil spray. Guide 02 and Guide 06.

PARTS AND REFERENCE · IAC, HOSES, TEST SCREENS

VERIFY ALL PART NUMBERS BY VIN

Item	Number	What the source actually says	Status
IAC motor, new	Mopar 4874373AB	Dealer catalog describes this as the idle-speed control motor for Jeep Grand Cherokee 1998–2004, and Wrangler 1998–2004. Aftermarket crosses such as Standard AC176 appear in listings. This PCM is reported to reject poor aftermarket units.	CATALOG LISTED Verify VIN. Buy new. Do not push the pintle.
PCV / CCV short hose	Dorman 47128 Chrysler 4854265	Dorman lists 47128 for 1997–2004 Grand Cherokee, including 4.0L. Two hoses, about 2.56 in, intake manifold to valve cover and MAP. OE cross 4854265. It is a small hose set, not the entire crankcase hose.	CATALOG LISTED Match it to the piece in your hand.
Vacuum elbow example	53030006	No Mopar vacuum-elbow application for this Jeep was found. A nearby number, 53030008, is a vacuum tee listed for a PT Cruiser, which is the wrong vehicle. Marketplace ads cite other elbow numbers. None of that is a confirmation.	NOT CONFIRMED Do not buy this number from this sheet.
HVAC / cruise vacuum line	55115901AG	Factory rapid-response 04-008, only for 2004 WJ 4.0L built 31 March–6 April 2004. Replaces a collapsing elbow and check valve at the intake, not the whole harness. That failure blocks HVAC and cruise vacuum. It does not cause a high idle.	ONLY IF VIN IS IN THAT WINDOW
Valve cover, filter-adapter O-rings, oil-pressure sender, pan gasket, rear main	Order by VIN after the leak is identified	No number is printed here. Buying a rear main for an adapter leak is the usual expensive miss on this engine.	TEST REQUIRED

Test screens used in this package

Test	Screen	How hard this number is
Warm idle, Park, loads off	About 700–800 rpm	Job screen. TSB 18-008-04 states about 744 rpm fully warm for domestic 2004 4.0L trucks built before 6 March 2004. CONFIRMED as the bulletin text. Not proof this VIN has that software.
IAC, each coil, isolated from the other	About 40–80 Ω. Field notes often 48–60 Ω.	LIKELY screen. Not a 2004 FSM page in hand. Good ohms do not save a pushed pintle.
O2 heater, sensor cold	About 4–10 Ω. Earlier 4.0L tests cite about 4–7 Ω.	LIKELY screen. Open or near zero is a failed heater. In-range means keep testing power and control.
O2 heater power	About 10–12 V while the feed is energized	Earlier 4.0L key-on tests use 10–12 V. ASD may drop out a second after key-on. TEST REQUIRED
Idle vacuum, near 700 ft	About 17–21 inHg, steady	LIKELY screen. Low or flickering has several causes.
IAC screws	7 N·m (60 in-lb) reported	LIKELY Do not strip the throttle body.
O2 sensor	22 ft-lb (30 N·m) in the WJ guide	LIKELY No extra anti-seize on a coated new sensor.

Brand note for this PCM. Oxygen sensors and IAC motors are the two parts this generation is repeatedly reported to be picky about. A new Mopar or NTK part, correct position, verified by VIN, is the purchase. A used IAC, a pushed pintle, or a universal oxygen sensor with cut wires is how the same codes come back.